



CS3219: Project Milestone 1 (D1)

Project Group 5

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Glossary

Accounts and access

Term	Meaning
User	Anyone with a registered account.
Account	A user's registered identity: an account identifier, username, email address, password, profile information, role and status.
Account Identifier	A unique, permanent identifier given to an account at registration and never reused.
Username	A unique name chosen at registration that identifies an account to other users. It cannot be changed.
Display Name	The name shown to other users alongside the username.
Profile Information	The details a user may edit about themselves: currently, their display name.
Account Role	Determines what an account may do. Every account has exactly one role: • MEMBER: given at registration; may create and accept errand requests. • ADMINISTRATOR: may also manage suppliers, change roles, suspend and restore accounts, and resolve disputes.
First Administrator	The first ADMINISTRATOR account, created by the deploying organisation. Later administrators are appointed by existing ones.
Institutional Domain	The email domain(s) allowed for registration, e.g., u.nus.edu.
Email Verification	The process by which a user proves they control an email address. A verification expires after 24 hours or once a newer one is requested.
Session	The period a user stays logged in, from login until logout, expiry, or the system ending it.
Audit Log	A permanent record of administrative actions.
Non-reversible Form	A stored version of a value (e.g., a password) from which the original cannot be recovered, but which can still be checked against a supplied value.

Errand requests

Term	Meaning
Errand Request	A request to collect items from a supplier and deliver them to a drop-off location. It has an identifier, requester, items, pickup and drop-off locations, credit value, listing, pickup and delivery deadlines, a current state, and a courier once assigned.
Requester	The user who created a given errand request. Fixed at creation.
Courier	The user currently assigned to fulfil a given errand request. An errand has at most one courier at a time. Being a courier is not an account role.
Active Errand Request	An errand request in a non-terminal state (see Errand Request States). Its credits are reserved.
Supplier	A store, facility or campus landmark where items can be collected.

Credits

Term	Meaning
Credits	Whole-number units used only within the platform. They cannot be bought, withdrawn or exchanged.
Reserved Credits	Credits held for an active errand request and unavailable for anything else.
Available Balance	A user's total credits minus their reserved credits.
Settlement	Moving an errand's reserved credits from the requester to the courier.
Release	Returning an errand's reserved credits to the requester's available balance.

Communication between services

Term	Meaning
Deploying Organisation	The organisation that installs and runs the platform, e.g., the university.
Account Status Notification	A message the User Service sends to other services when an account first becomes ACTIVE, or is suspended, restored or closed.
Operation Identifier	A unique identifier for one operation, kept the same if the operation is retried, so that it is never applied twice.

Miscellaneous

Normal Operating Load	We assume a deployment of around 15,000 (out of around 45,000 NUS students) registered users of whom at most 1000 are concurrently active during peak periods. We assume a peak of 20 new errands created per second and at most 80 errand state transitions per second.
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Errand Request States

State	Meaning	Terminal
OPEN	Visible to couriers; credits reserved.	No
ACCEPTED	Assigned to a courier.	No
PICKED_UP	Items collected by the courier.	No
DELIVERED	Courier reported delivery; awaiting requester confirmation.	No
DISPUTED	Delivery reported but rejected by the requester; awaiting administrator decision	No
COMPLETED	Delivery confirmed; credits settled to the courier.	Yes
CANCELLED	Cancelled by the requester before pickup; credits released.	Yes
EXPIRED	Not accepted before the listing deadline; credits released.	Yes
FAILED	Not delivered, or dispute decided for the requester; credits	Yes

Account States

State	Meaning	Terminal
PENDING_VERIFICATION	Email not yet verified; cannot log in. Discarded after 24 hours.	No
ACTIVE	Email verified; can log in.	No
SUSPENDED	Suspended by an administrator; cannot log in.	No
CLOSED	Closed by the user; cannot log in or be reopened.	Yes

Planned Sprint / Iteration

Sprint / Iteration Number	Milestone
1	Milestone D2 – Progress Check 1
2	Milestone D3 – Progress Check 2
3	Milestone D4 – Final Submission and Presentation

Product Backlog

1. Functional Requirements

User Service			Priority	Planned Sprint/iteration
US-F1	Support account registration.			
	US-F1.1	The system shall allow a person to register an account with a username, an email address and a password.	High	1
	US-F1.1.1	The system shall reject a username already used by another account, including a closed account.	High	1
	US-F1.1.2	The system shall reject a username shorter than 3 or longer than 30 characters, or containing characters other than ASCII letters, digits and underscores.	High	1
	US-F1.1.3	The system shall reject a password shorter than 12 characters or lacking at least one uppercase letter, one lowercase letter and one digit.	High	1
	US-F1.2	The system shall allow a MEMBER to act as both a requester and a courier.	High	1
US-F2	Restrict accounts to verified members of the institution.			
	US-F2.1	The system shall allow registration only with an email address in the institutional domain.	High	1
	US-F2.1.1	The system shall reject a registration whose email address is outside the institutional domain.	High	1
	US-F2.1.2	The system shall treat an email address as inside the institutional domain only if everything after the "@" matches a configured domain, ignoring letter case.	High	1
	US-F2.2	The system shall require an account's email address to be verified before the account can be used.	High	1

User Service			Priority	Planned Sprint/iteration
	US-F2.2.1	The system shall create every new account with the status PENDING_VERIFICATION.	High	1
	US-F2.2.2	The system shall change an account's status to ACTIVE when its email address is verified.	High	1
	US-F2.2.3	The system shall allow a PENDING_VERIFICATION account to request a new verification, after which earlier verifications no longer work.	Medium	2
	US-F2.2.4	The system shall discard an account that is still PENDING_VERIFICATION 24 hours after creation, making its username and email address available again.	Medium	2
	US-F2.3	The system shall allow at most one account per verified email address.	High	1
	US-F2.3.1	The system shall reject a registration whose email address matches that of an existing account, ignoring letter case.	High	1
	US-F2.3.2	The system shall reject a registration whose email address was ever verified for another account, including a closed account or an address that account has since replaced.	High	1
US-F3	Support logging in and keeping accounts secure.			
	US-F3.1	The system shall log in a user who supplies the username or email address of an ACTIVE account together with its password.	High	1
	US-F3.1.1	The system shall reject invalid login details without revealing which field was wrong.	High	1
	US-F3.1.2	When the correct password is supplied for an account that is not ACTIVE, the system shall reject	High	1

User Service			Priority	Planned Sprint/iteration
		the login and state whether the account is pending verification or suspended.		
	US-F3.2	The system shall allow a logged-in user to log out, after which requests from that session are no longer accepted.	Medium	2
	US-F3.3	The system shall end a session after 30 minutes without activity, and in any case 24 hours after login.	Medium	2
	US-F3.4	The system shall allow a logged-in user to change their password after entering their current password.	Medium	2
	US-F3.4.1	The system shall end all of the user's other sessions after a password change.	Medium	2
	US-F3.5	The system shall allow a user who has forgotten their password to set a new one after proving they control the account's email address.	Medium	2
	US-F3.5.1	The system shall respond to a password reset request in the same way whether or not the email address belongs to an account.	Medium	2
	US-F3.5.2	The system shall end all of the account's sessions after a password reset.	Medium	2
	US-F3.6	The system shall apply the rules in US-F1.1.3 to every new password.	Medium	2
US-F4	Support account roles so that administrative operations are limited to administrators.			
	US-F4.1	The system shall give every account exactly one role: MEMBER or ADMINISTRATOR.	High	1

User Service			Priority	Planned Sprint/iteration
	US-F4.1.1	The system shall give the MEMBER role to every account created through registration.	High	1
	US-F4.1.2	The system shall allow only an ADMINISTRATOR to change another account's role.	High	1
	US-F4.1.3	The system shall reject a role change that would leave the platform with no ACTIVE administrator.	High	1
	US-F4.1.4	A role change shall take effect immediately, including for users who are already logged in.	High	1
US-F5	Allow the platform to be handed over to the deploying organisation without any hard-coded administrator access.			
	US-F5.1	The system shall contain no administrator accounts or hard-coded administrator passwords when first deployed.	High	1
	US-F5.2	The system shall allow only the deploying organisation to create the first administrator account.		
	US-F5.3	The system shall permanently disable first-administrator creation once any administrator account exists.	High	1
US-F6	Let other services identify who is making a request, so that each service decides its own permissions.			
	US-F6.1	The system shall let any service confirm whether a request comes from a logged-in user and, if so, obtain that user's account identifier and role.	High	1
	US-F6.2	The User Service shall decide permissions only for operations on accounts, roles and profiles.	High	1
	US-F6.3	The system shall not store "requester" or "courier" as an account role.	High	1

User Service			Priority	Planned Sprint/iteration
US-F7	Support profile management.			
	US-F7.1	The system shall allow a logged-in user to view and update their own profile information.	High	1
	US-F7.1.1	The system shall reject an attempt to update another account's profile information.	High	1
	US-F7.1.2	The system shall reject a display name that is empty or longer than 50 characters.	High	1
	US-F7.1.3	The system shall not allow a username to be changed after registration.	Medium	2
	US-F7.1.4	The system shall allow a logged-in user to see another account's username and display name, but not its email address.	Medium	2
	US-F7.2	The system shall allow a logged-in user to change their email address after entering their current password.	Medium	2
	US-F7.2.1	The system shall apply the rules in US-F2.1 and US-F2.3 to the new email address.	Medium	2
	US-F7.2.2	The system shall keep the current email address until the new one is verified, and keep it if verification is not completed.	Medium	2
US-F8	Support account closure and suspension.			
	US-F8.1	The system shall allow a user to close their own account after entering their current password.	Medium	2
	US-F8.1.1	The system shall reject a closure while the account is the requester or courier of any active errand request.	Medium	2

User Service			Priority	Planned Sprint/iteration
	US-F8.1.2	The system shall end all sessions of an account when it is closed.	Medium	2
	US-F8.1.3	The system shall not allow a closed account to be reopened.	Medium	2
	US-F8.1.4	For a closed account, the system shall keep only its account identifier, its username, and a non-reversible record of each email address verified for it, enough to enforce US-F1.1.1 and US-F2.3.2.	Medium	2
	US-F8.1.5	The system shall delete a closed account's password and profile information.	Medium	2
	US-F8.2	The system shall allow an ADMINISTRATOR to suspend any other account, including another administrator's.	Medium	2
	US-F8.2.1	The system shall end all sessions of an account when it is suspended.	Medium	2
	US-F8.2.2	The system shall allow an ADMINISTRATOR to restore a suspended account to ACTIVE.	Medium	2
	US-F8.2.3	The system shall reject an administrator's attempt to suspend their own account.	Medium	2
US-F9	Inform other services of account status changes, so that they can react without the User Service knowing their rules.			
	US-F9.1	The system shall notify other services when an account becomes ACTIVE for the first time.	High	1
	US-F9.2	The system shall notify other services when an account is suspended, restored or closed.	Medium	2

User Service			Priority	Planned Sprint/iteration
	US-F9.3	Each account status notification shall include the account identifier, the new status, the time of the change and an operation identifier.	Medium	2

Supplier Service			Priority	Planned Sprint/iteration
SS-F1		Support viewing supplier information via the supplier panel.		
	SS- F1.1	The system should support viewing supplier information when an authenticated user logs in.	High	1
	SS- F1.1.1	The system should display a list of available stores (e.g., CoffeeBean@Com3), facilities (e.g., Printers@PCCommons), and/or locations (e.g., PGP Foyer) from which authenticated users may request errands.	High	1
	SS- F1.1.2	The system should display the supplier's details, such as name, description, and opening hours.	High	1
SS- F2		Support managing supplier information via the supplier panel		1
	SS- F2.1	The system should allow an administrator to create a supplier or pickup-location record.	High	1
	SS- F2.1.1	The system should require a name, category, description, and location before creating a supplier record.	High	1
	SS- F2.1.2	The system should save a valid supplier record to a relational database and make it available to supplier listings.	High	1

	SS-F2.1.3	The system should reject an incomplete or invalid record and display an appropriate error message to the administrator.	High	1
	SS- F2.2	The system should allow an administrator to delete a supplier or pickup-location record.	High	1
	SS-F2.2.1	The system should require the administrator to select an existing record before requesting its deletion.	Medium	1
	SS-F2.2.2	The system should request administrator confirmation before deleting the selected record.	Medium	1
	SS-F2.2.3	The system should prevent a record associated with an active errand request from being deleted.	High	1
	SS-F2.2.4	The system should remove a successfully deleted record from the supplier listing and prevent it from being selected for new errand requests.	High	1
	SS-F2.2.5	The system should display whether the deletion was successful or unsuccessful (with a clear, actionable error message).	High	1
	SS- F2.3	The system should allow an administrator to update information regarding a supplier or pickup-location record.	High	1
	SS-F2.3.1	The system should allow the administrator to update the record name, category, description, and location.	High	1
	SS-F2.3.2	The system should require all mandatory fields to remain present after an update.	High	1
	SS-F2.3.3	The system should validate the updated information to the team-defined field constraints.	High	1

	SS-F2.3.4	The system should save valid changes and display the updated information in the supplier listing.	High	1
	SS-F2.3.5	The system should reject invalid changes and retain the previously saved information (with a clear, actionable error message).	Medium	1
	SS-F2.3.6	The system should detect concurrent updates to the same supplier record, accept one valid update, and reject any stale conflicting update.	High	1

Order Service			Priority	Planned Sprint/iteration
OS-F1	The Order Service shall allow a user to create a new errand request, which enters the OPEN state and reserves the requester's credits.			
	OS- F1.1	On creation, the service shall record the errand details (requestor, items, pickup and drop-off locations, and credit value) and set its state to OPEN.	High	2
	OS-F1.1.1	On creation, the service shall reserve the requester's credits equal to the errand's value, making the errand visible to couriers as an open request.	High	2
	OS-F1.2	If the Credit Service declines the reservation of credits upon creation, the service shall reject the creation of the errand	High	2

	OS-F1.3	On creation, the service shall require a listing, pickup, and delivery deadline that are in the future and ordered (listing deadline $\leq$ pickup deadline $\leq$ delivery deadline)	Medium	3
OS-F2		The Order Service shall bring every errand to a terminal state that reflects the lifecycle phase in which it ended and shall settle the errand's reserved credits when that terminal state is reached.		
	OS- F2.1	The service shall resolve errands that end while still OPEN, before any courier has been assigned.	High	2
	OS- F2.1.1	When the requester cancels an errand that is still OPEN, the service shall transition it to CANCELLED and release the reserved credits back to the requester.	High	2
	OS- F2.1.2	When an OPEN errand is not accepted before its listing deadline, the service shall transition it to EXPIRED and release the reserved credits back to the requester.	High	2
	OS- F2.2	The service shall resolve errands that have been ACCEPTED but not yet picked up.	High	2
	OS- F2.2.1	When the requester cancels an errand that is ACCEPTED, the service shall transition it to CANCELLED and release the reserved credits back to the requester.	Medium	3
	OS- F2.2.2	When the assigned courier withdraws, or the pickup deadline passes before the errand is marked PICKED_UP, the service shall return the errand to OPEN so that it can be re-listed for another courier.	Medium	3
	OS- F2.3	The service shall resolve errands that have been PICKED_UP but not delivered.	High	2

	OS-F2.3.1	When the delivery deadline passes before an errand is marked DELIVERED, the service shall transition it to DISPUTED and hold the reserved credits in pending resolution by an admin.	Medium	3
	OS- F2.4	The service shall resolve errands that have been marked DELIVERED by the courier.	High	2
	OS-F2.4.1	When the requester rejects a DELIVERED errand, the service shall transition it to DISPUTED and hold the reserved credits in pending resolution by an admin.	Medium	3
	OS- F2.5	The service shall resolve errands that are DISPUTED through an admin.	High	2
	OS-F2.5.1	When an admin resolves a DISPUTED errand in favor of the courier, the service shall transition it to COMPLETED and transfer the reserved credits to the courier.	High	2
	OS-F2.5.2	When an admin resolves a DISPUTED errand in favor of the requester, the service shall transition it to FAILED and release the reserved credits back to the requester.	High	2
OS- F3		The Order Service shall advance an errand through its lifecycle as the requester and courier perform each step.		
	OS- F3.1	When a courier accepts an OPEN errand, the service shall transition it to ACCEPTED and assign that courier.	High	2
	OS-F3.1.1	The service shall assign at most one courier to an errand. Once an errand is ACCEPTED, no further accept shall succeed until the errand returns to OPEN (OS-F2.2.2).	High	2

	OS-F3.1.2	The service shall reject an accept made by the errand's own requester, leaving the errand in the OPEN state.	High	2
	OS-F3.1.3	When two or more users attempt to accept the same OPEN errand, the service shall assign exactly one courier and reject every other attempt with a result indicating the errand is no longer available.	High	2
	OS-F3.2	When the assigned courier marks an ACCEPTED errand as collected, the service shall transition it to PICKED_UP.	High	2
	OS-F3.3	When the assigned courier marks a PICKED_UP errand as delivered, the service shall transition it to DELIVERED.	High	2
	OS-F3.4	When the requester confirms receipt of a DELIVERED errand, the service shall transition it to COMPLETED and transfer the reserved credits to the courier.	High	2
	OS-F3.5	When the requester does not confirm or reject a DELIVERED errand before the confirmation deadline, the service shall transition it to COMPLETED and settle the reserved credits to the courier.	Medium	3
OS-F4	The Order Service shall allow a user to browse the errand requests that are currently available to accept.			
	OS-F4.1	The service shall let a logged-in user view the list of errand requests that are in the OPEN state.	High	2
	OS-F4.1.1	The service shall exclude from this list any errand that is created by the requesting user	High	2
	OS-F4.1.2	For each listed errand, the service shall show the information relevant to the errand, the supplier	High	2

		location, drop-off location, requested items, and any relevant deadlines.		
	OS-F4.1.3	The service shall only list errands that are in the OPEN state	High	2
OS-F5	On reaching a terminal state, the Order service shall publish a terminal event to the Credit Service so that the errand's reserved credits are settled or released back to the user who created the errand			
	OS-F5.1	The service shall publish exactly one terminal event when an errand transitions to COMPLETED, CANCELLED, EXPIRED or FAILED	High	2
	OS-F5.1.1	The service shall record the terminal state change and enqueue its event in a single atomic operation	High	2
	OS-F5.1.2	Each event shall carry an unique operation identifier, the errand, the event type (OrderCompleted, OrderCancelled, OrderExpired or OrderFailed), the requester, the courier if any and the credit amount	High	2
	OS-F5.2	Event delivery shall be at-least-once and persistent, meaning a published event should not be lost	High	2
OS-F6	The order service shall react to account suspensions that affect active errand by consuming account-status notifications from the User Service (US-F9.2)			
	OS-F6.1	Upon receiving a notification that a user has been suspended, the service shall resolve every active errand in which that user is the requester or the assigned courier.	Medium	3

	OS-F6.1.1	When the errand is OPEN, the service shall transition it to CANCELLED and release the reserved credits to the requester.	Medium	3
	OS-F6.1.2	When the errand is ACCEPTED and the suspended participant is its assigned courier, the service shall unassign the courier and return the errand to OPEN.	Medium	3
	OS-F6.1.3	In every other active state, the service shall route the errand to DISPUTED for administrator resolution.	Medium	3
	OS-F6.2	The service shall process each account-status notification at most once, identified by its operation id	Medium	3
OS-F7	The Order Service shall allow a user to view errands they are involved in, as a requester or a courier			
	OS-F7.1	The service shall let a user list the errands they created, showing each errands current errand state	High	2
	OS-F7.2	The service shall let a user list the errands they are assigned to as courier, showing each errands current errand state	Medium	3

Credit Service			Priority	Planned Sprint/iteration
CS-F1	Support a closed credit economy on the platform. (M5)			
	CS- F1.1	The system shall allocate 100 credits when an account first becomes ACTIVE, following the account-status notification from User Service.	High	1
	CS- F1.1.1	The system should allocate the initial credits exactly once per account.	High	1
	CS- F1.2	The system shall allow an authenticated user to view their own credit balances.	High	1
	CS- F1.2.2	The system shall display the user's total balance, reserved credits and available balance.	High	
	CS- F1.2.3	The displayed available balance shall equal total balance minus reserved credits.	High	
	CS- F1.2.4	The system shall reject an attempt to view another user's private credit balances.	High	
	CS- F1.3	The system should not provide any mechanism by which credits can be purchased, withdrawn, or exchanged for external value.	High	1
	CS- F1.4	Credits shall be represented as whole-number units. The system shall reject fractional operation amounts.	High	
CS-F2	Reserve settle and release errand credits (M5)			
	CS- F2.1	On a reservation request from Order Service, Credit Service shall reserve the offered amount only if it is a positive whole number within the requester's available balance, and return success or rejection	High	1
	CS- F2.1.1	A successful reservation shall increase reserved credits and reduce available balance equally, leaving total balance unchanged. Order Service	High	1

Credit Service			Priority	Planned Sprint/iteration
		shall publish the errand as OPEN only after success.		
	CS-F2.1.2	Each errand shall have at most one reservation. Credits shall remain reserved throughout OPEN, ACCEPTED, PICKED_UP, DELIVERED and DISPUTED	High	1
	CS-F2.1.3	Returning ACCEPTED to OPEN after courier withdrawal, pickup timeout or courier suspension shall preserve the existing reservation and amount without reserving credits again.	High	
	CS-F2.2	On receiving an OrderCompleted event from Order Service under OS-F5, Credit Service shall transfer the reserved amount from the requester to the courier identified in that event.	High	
	CS-F2.2.1	Settlement of X credits shall reduce the requester's reserved balances by X, increase the courier's total balance by X, and finalize the reservation as settled.	High	
	CS-F2.2.2	An OrderCancelled, OrderExpired or OrderFailed event shall release the reservation: reduce reserved credits by the held amount, leave total balance unchanged and pay no courier.	High	1
	CS-F2.3	Entering DISPUTED shall retain the reservation. Settlement shall occur only on OrderCompleted, release shall occur on OrderFailed following resolution for the requester	High	
	CS-F2.3.1	OrderCompleted shall have the same credit effect after requester confirmation (OS-F3.4), confirmation timeout (OS-F3.5), or dispute resolution for the courier (OS-F2.5.1)	High	
	CS-F2.4	Credit Service shall accept terminal credit events only from Order Service. The event type shall be	High	

Credit Service			Priority	Planned Sprint/iteration
		one of the four types in OS-F5.1.2. Its errand, requester and amount shall match the existing reservation.		
	CS-F2.4.1	OrderCompleted shall identify the courier recorded by Order Service at completion, different from the requester. Invalid events shall produce a rejection and no balance changes	High	
	CS-F2.5	Duplicate deliveries of the same terminal event shall cause one set of credit changes and one successful transaction record. A settled reservation shall not be released and a released reservation shall not be settled	High	
CS-F3	Maintain credit transaction history. (M5)			
	CS- F3.1	The system should record every credit transaction with a unique transaction identifier, type, amount, accounts identifier, balance changes and timestamp.	High	2
	CS-F3.1.1	The system shall provide no operation for modifying or deleting an existing credit transaction record.	High	2
	CS-F3.1.2	The system shall allow a logged-in user to retrieve only transactions affecting their own account, ordered newest first by timestamp and then transaction identifier.	Medium	3
CS-F4	Handle credits across the account lifecycle (M5)			
	CS- F4.1	Suspension or restoration notified under US-F9.2 shall not itself change credits.	Medium	2
	CS-F4.1.1	Terminal events for existing errands shall still be processed during suspension.	Medium	

Credit Service			Priority	Planned Sprint/iteration
	CS- F4.2	Following a CLOSED notification under US-F9.2, Credit Service shall forfeit the remaining balance after all earlier errand-related credit operations are resolved and no credits remain reserved	Medium	2
	CS- F4.2.1	Forfeiture shall reduce total and available balances to zero and record the positive amount with the closing account as source and no destination.	Medium	2
	CS- F4.2.2	A zero balance requires no forfeiture transaction.	Medium	

This table summarizes the asynchronous workflows required by M6. The requirements for each workflow appear in the tables of the services that send and receive the notifications.

Asynchronous Workflows	Sent by	Received By
Initial Credit Allocation	US-F9.1	CS-F1.1
Credit settlement and release	OS-F5	CS-F2
Balance forfeiture on closure	US-F9.2	CS-F4.2
Handling a suspended courier's errands	US-F9.2	OS6

Note: The entries in the planned sprint/iteration column in the above table should be aligned with the project plan. While we have not asked to develop a project plan explicitly, you will still need a project plan. Include it as a separate section within this product backlog. You

can choose any format for the project plan, for example, a Gantt chart to show the overall project plan.

User Service			Priority	Planned Sprint/iteration
US-NFR1	Security - Auditing			
	US-NFR1.1	The system shall record every role change, suspension, restoration and first-administrator creation in an audit log.	High	2
	US-NFR1.1.1	Each audit entry shall record the acting administrator, the target account, the action, the previous and new values, and the time.	High	2
	US-NFR1.1.2	An administrative action shall take effect only if its audit entry is saved; otherwise the action shall be rejected and nothing shall change.	High	2
	US-NFR1.2	The system shall provide no way for any user, including administrators, to edit or delete an audit entry	Medium	2
	US-NFR1.2.1	Any change made to a stored audit entry shall be detectable when the audit log is checked.	Medium	2
US-NFR2	Security – Credential confidentiality.			
	US-NFR2.1	The system shall store passwords only in a form that cannot be converted back into the password.	High	1
	US-NFR2.2	The system shall never include a password in any response, log or audit entry.	High	1
	US-NFR2.3	The system shall accept login details only over encrypted connections.	High	1

Order Service			Priority	Planned Sprint/iteration
OS-NFR1	Timeliness – Dispute observability			

	OS-NFR1.1	For at least 95% of disputes under normal load, an admin shall be able to observe an errand as DISPUTED within 1 second of the dispute being triggered	Medium	3
OS-NFR2	Integrity – Concurrency safe transitions			
	OS-NFR2.1	Under any number of simultaneous accepts on one OPEN errand, exactly one shall be assigned. The invariant of OS-F3.1.1 and OS-F3.1.3 shall hold regardless of concurrency.	High	2
OS-NFR3	Reliability – Durable terminal event emission			
	OS-NFR3.1	A terminal transition and its outbound settlement event shall persist as one atomic unit. A committed terminal state shall always result in exactly one effective settlement event, even if the service restarts immediately after committing.	High	2
	OS-NFR3.2	No committed terminal transition shall leave its reserved credits unsettled once the Credit Service has processed the event.	Medium	3

Supplier Service		Priority	Planned Sprint/iteration
SS-NFR1	Performance - Responsiveness		
	SS-NFR1.1	The Supplier Service should respond promptly when supplier information is requested.	
	SS-NFR1.1.1	At least 95% of supplier-list and supplier-detail requests should be completed within 2 seconds under the team-defined expected workload.	
SS-NFR2	Security - Authorization		

	SS-NFR2.1	The system should restrict supplier-management operations to administrators.		
	SS-NFR2.1.1	All unauthorized create, update, and delete requests should be rejected without modifying supplier information.		
SS-NFR3	Integrity – Concurrent Updates			
	SS-NFR3.1	The system should maintain complete and consistent supplier information when multiple administrators update the same record concurrently.		
	SS-NFR3.1.1	The system should accept one valid update and reject any stale conflicting update without partially overwriting the supplier's record.		

Credit Service			Priority	Planned Sprint/iteration
CS-NFR1	Transaction atomicity			
	CS-NFR1.1	Each credit transaction shall take effect in full or have no effect. The system shall not expose or retain a partially applied transaction.	High	
	CS-NFR1.1.1	For settlement of an amount X, the requester's reserved balances shall each decrease by X, and the courier's total balance shall increase by X. These changes shall constitute one indivisible outcome.	High	
	CS-NFR1.2	If a credit transaction fails before taking effect in full, its changes must be reverted	High	
CS-NFR2	Duplicate delivery and recovery			
	CS-NFR2.1	For each valid logical credit operation, repeated delivery shall apply its credit effects and successful transaction record at most once, including after restart.	High	

	CS-NFR2.1.1	Delivering the same terminal event 20 times, including concurrently and after restart following success but before its result is received, shall produce one finalisation and one successful record	High	
	CS-NFR2.2	Successfully completed balances, reservation outcomes, transaction records and operation results shall survive service and data store restart with persistent storage intact	High	
	CS-NFR2.3	A missing response shall not be treated as proof of failure. Retry or result retrieval shall return the existing outcome without reversing or repeating a completed transaction	High	
CS-NFR3	Credit consistency and concurrency			
	CS-NFR3.1	Every committed balance shall satisfy total $\geq$ reserved $\geq$ 0 and available = total - reserved. Reserved credits shall equal the sum of that account's unresolved reservations	High	
	CS-NFR3.2	Concurrent operations shall not cause overspending, lost updates or both settlement and release of one reservation. Balance responses shall use one committed state.	High	
	CS-NFR3.2.1	Concurrent valid settlements from different errands to one courier shall retain every successful credit increase, conflicting finalizations of one reservation shall allow at most one to succeed.	High	

2. Selected Nice-to-haves

N1	Credit Transaction Search and Filters			
	N1-FR1	The system shall allow users to search their own credit transaction history by transaction or errand identifier and filter by transaction type, date range and credit amount range.		

N2	Natural Language Errand Creation			
	N2-FR1	The system should allow the requester to create an errand using natural language.		
	N2-FR1.1	The system should extract relevant errand information from the user's natural language prompt.		
	N2-FR1.1.1	The extracted information via an external LLM model should be used to populate the relevant mandatory fields for errand creation such as supplier location, delivery location, and deadlines.		

N3	Observability – Centralized Logging			
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	N3-FR1	Every service shall emit structured logs and ship them to a central logger, so that events can be traced across services.		
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3. UI Mockup/screenshots

Link to UI Mockup (Hosted on ChatGPT Sites): <https://campuserrand-group5-mockup.ron52086.chatgpt.site/>

Courier Workflow

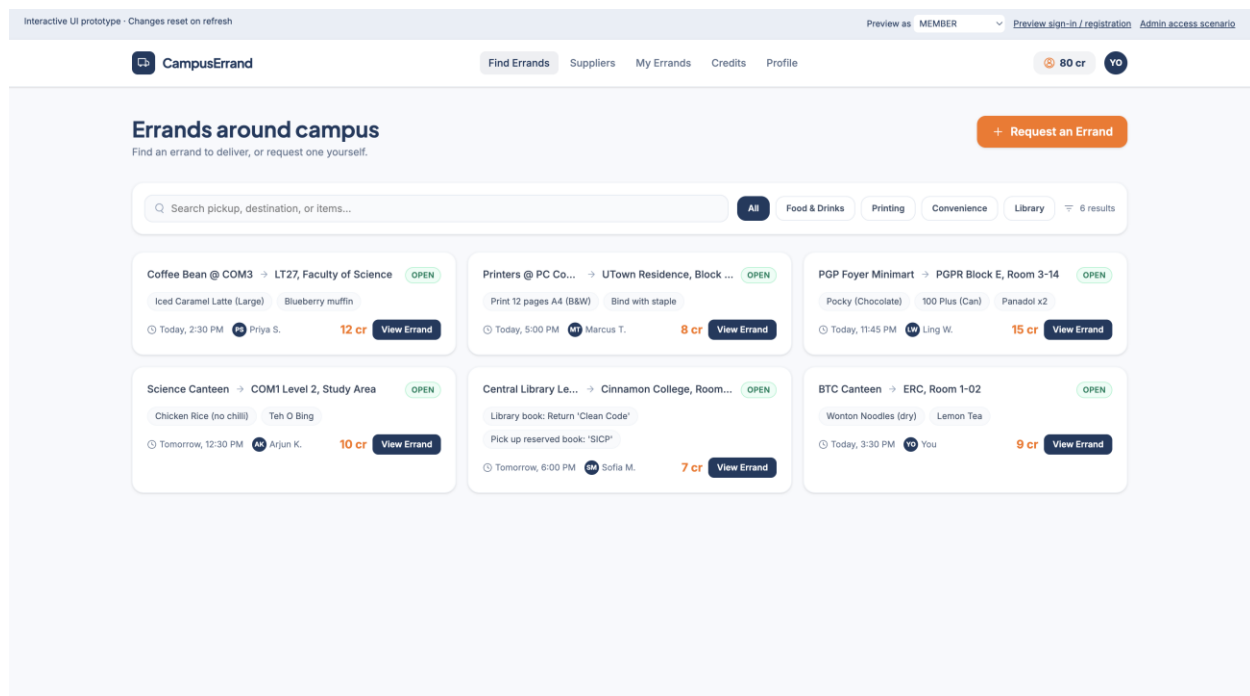


Figure 1: Page displaying list of open/available errands

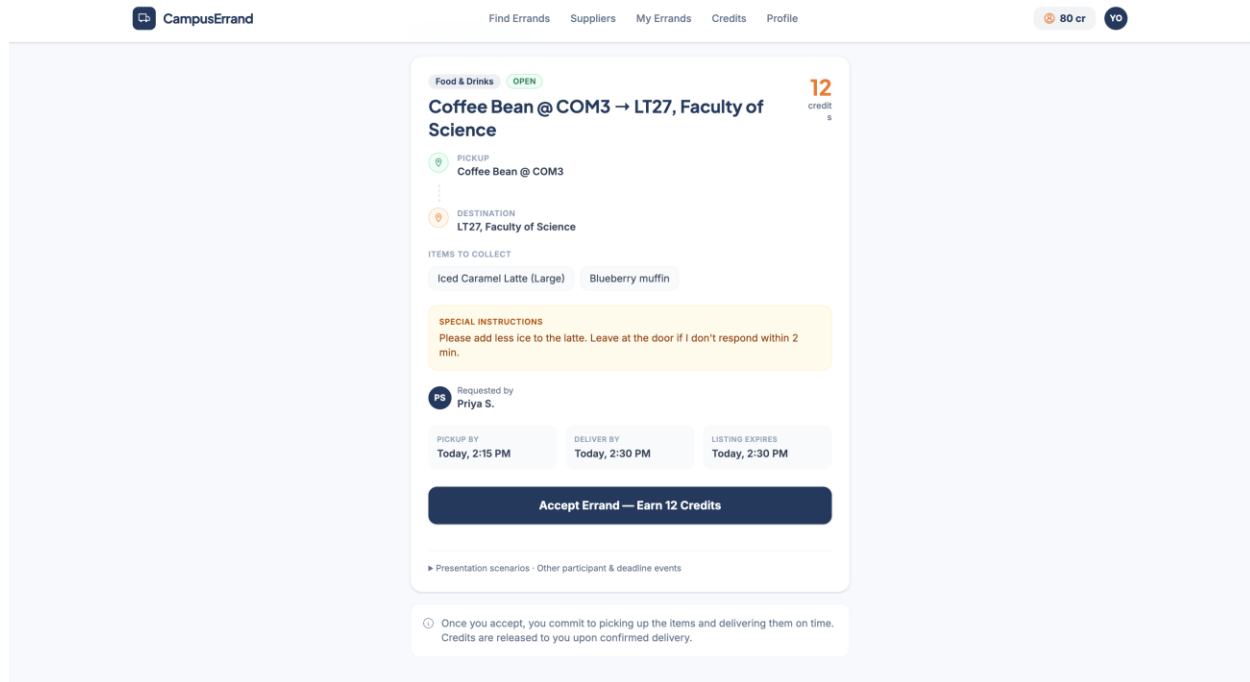


Figure 2: Page for viewing and accepting open/available errands

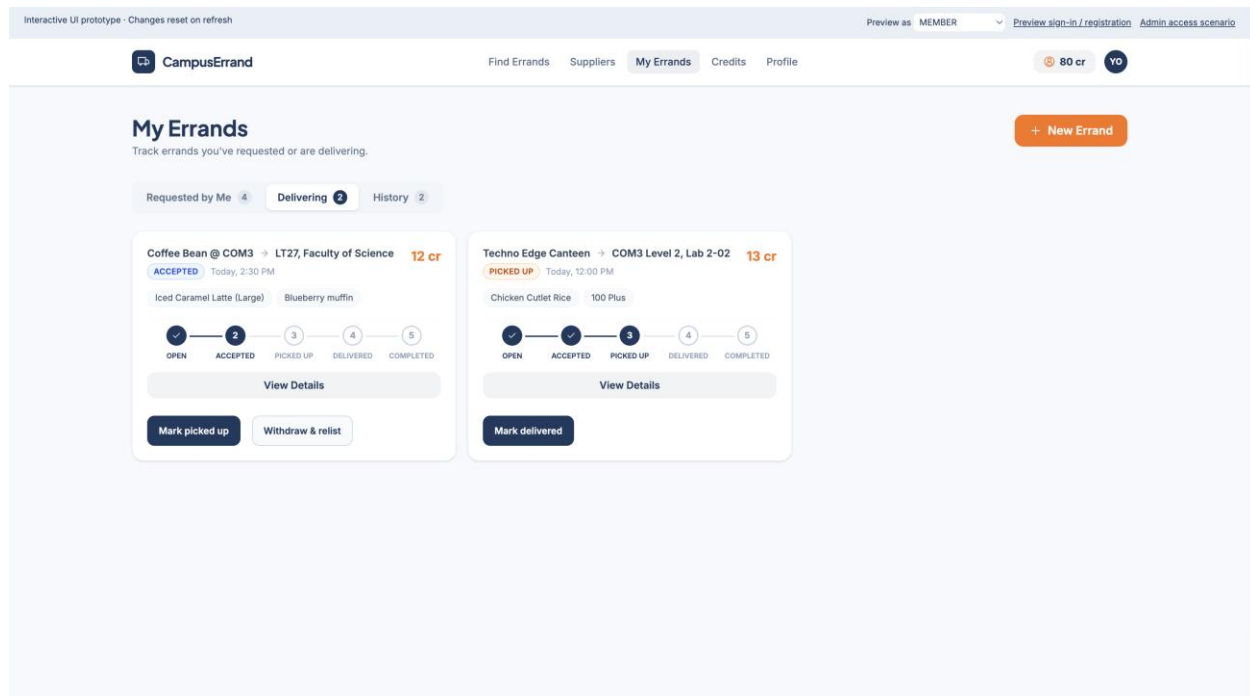


Figure 3: Page for tracking and updating states of errands

Requester Workflow

The screenshot shows the 'Request an Errand' form in the CampusErrand application. The header includes the CampusErrand logo, navigation links (Find Errands, Suppliers, My Errands, Credits, Profile), and a user profile with 80 credits and a 'YO' status. The form itself is titled 'Request an Errand' with a subtext 'Describe what you need and offer credits to a courier.' It includes a balance indicator showing 80 credits, with 40 reserved. The form fields are:

- Supplier / Pickup Location ***: A text input field with a location pin icon and placeholder text 'e.g. Coffee Bean @ COM3'.
- Items to Collect ***: A text area with placeholder text 'List each item on a new line e.g. Iced Caramel Latte (Large) Blueberry muffin'.
- Delivery Location ***: A text input field with a location pin icon and placeholder text 'e.g. COM1 Level 2, Study Area near window'.
- Special Instructions**: A text area with placeholder text 'e.g. Less ice, knock twice, leave at door...'.
- Credits Offered ***: A text input field with a coin icon and placeholder text 'e.g. 12'.
- Deadlines**: Three date pickers for 'Listing Deadline', 'Pickup Deadline', and 'Delivery Deadline *', all with the format 'dd/mm/yyyy, --:-- --'.

 At the bottom of the form are two buttons: 'Post Errand' (orange) and 'Cancel' (white).

Figure 1: Page for Requester to fill in relevant details to create a new errand

The screenshot shows the 'My Errands' page in the CampusErrand application. The header is identical to Figure 1. The page title is 'My Errands' with a subtext 'Track errands you've requested or are delivering.' There is a '+ New Errand' button in the top right. Below the title, there are three tabs: 'Requested by Me' (4), 'Delivering' (2), and 'History' (2). The main content area displays three errand cards:

- Frontier Canteen → YIH Level 3, Room 307**: Status 'ACCEPTED', time 'Today, 1:00 PM'. Items: Nasi Lemak, Milo Dinosaur. Progress bar: 1 (OPEN) → 2 (ACCEPTED) → 3 (PICKED UP) → 4 (DELIVERED) → 5 (COMPLETED). Buttons: 'View Details', 'Cancel errand'.
- BTC Canteen → ERC, Room 1-02**: Status 'OPEN', time 'Today, 3:30 PM'. Items: Wonton Noodles (dry), Lemon Tea. Progress bar: 1 (OPEN) → 2 (ACCEPTED) → 3 (PICKED UP) → 4 (DELIVERED) → 5 (COMPLETED). Buttons: 'View Details', 'Cancel errand'.
- Frontier Canteen → YIH Level 3, Room 307**: Status 'DELIVERED', time 'Today, 1:00 PM'. Items: Nasi Lemak, Milo Dinosaur. Buttons: 'View Details', 'Confirm receipt & pay 11 credits', 'Reject delivery / dispute'. A note at the bottom says 'Awaiting requester confirmation. Credits remain reserved.'

 At the bottom, there is another errand card:

- BTC Canteen → ERC, Room 1-02**: Status 'DISPUTED', time 'Today, 3:30 PM'. Items: Wonton Noodles (dry), Lemon Tea. Button: 'View Details'. A note at the bottom says 'Credits held while an administrator reviews this dispute.'

Figure 2: Page to view and track errands created by Requester

Authentication Workflow

Interactive UI prototype - Changes reset on refresh

Preview as: MEMBER [Preview sign-in / registration](#) [Admin access scenario](#)

CampusErrand - Your campus, connected

Welcome back

Request a helping hand or earn credits by delivering an errand.

[Sign in](#) [Create account](#)

Sign in to CampusErrand

Username or email

Password

[Sign in](#)

► Presentation scenarios & demo credentials

Figure 1: Sign-In page (Mobile View)

Interactive UI prototype - Changes reset on refresh

Preview as: MEMBER [Preview sign-in / registration](#) [Admin access scenario](#)

CampusErrand - Your campus, connected

Join your campus community

Request a helping hand or earn credits by delivering an errand.

[Sign in](#) [Create account](#)

One account. Request and deliver.

Username

Institutional email

Password

Username: 3-30 letters, digits or underscores. Password: 12+ characters with uppercase, lowercase and a digit. Institutional domain: u.nus.edu.

[Create account - Get 100 credits](#)

► Presentation scenarios & demo credentials

Figure 2: Account creation page

Administrator Workflow

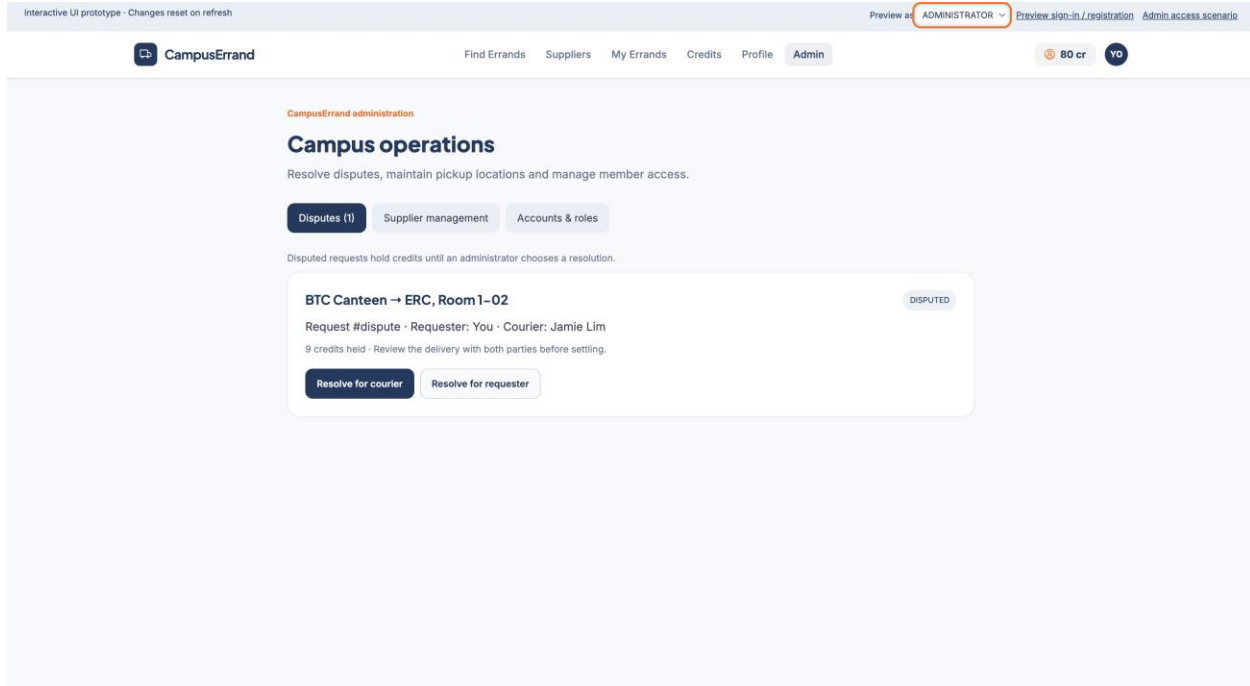


Figure 1: Dispute resolution page

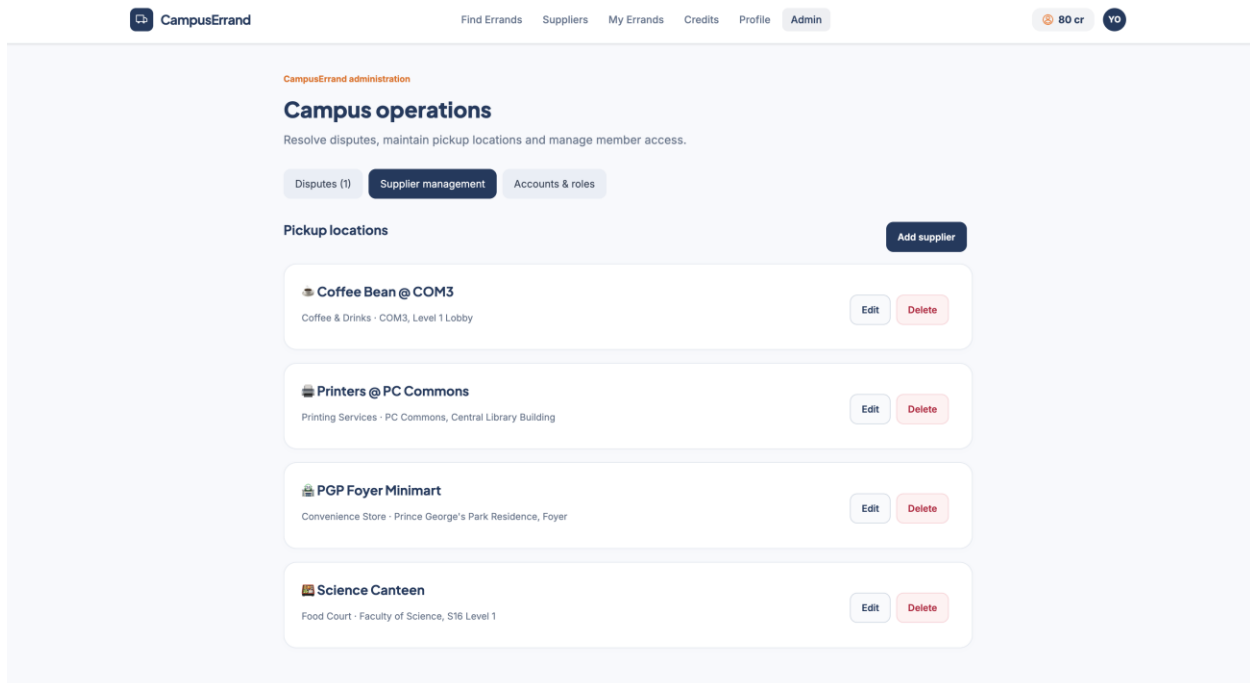


Figure 2: Page for Administrators to manage supplier information

User Workflow (Requester / Courier Agnostic)

CampusErrand • Your campus, connected

Your account
Manage your profile, email and account.

Profile information ADMINISTRATOR - ACTIVE

Username
campus_member
Your username cannot be changed.

Display name
Campus Member

Email: member@u.nus.edu

[Save profile](#) [Sign out](#)

Other member profile
Alex Tan - @alex_tan
Only username and display name are shared. Email is private. Only the account holder can edit this profile.

Change email address
Current password

Figure 1: Account information page

Credits
Earn credits by delivering errands. Spend them by requesting errands.

Total Balance
109
Current credits, including reserved credits

Reserved
29
Held for unsettled errands, including disputes

Available
80
Ready to spend on new errands

Total Earned
+138 cr

Total Spent
-29 cr

Every new account receives 100 credits once. Earn more by completing deliveries. Credits cannot be purchased, withdrawn, or exchanged for external value.

Transaction History

	All	Earned	Spent
COMPLETED - Errand awaiting - You → Alex Tan 9/17/2020, 4:58:56 PM			-11 cr
You → reservation - Errand dispute Today, 11:30 AM			-9 cr
You → reservation - Errand awaiting Today, 10:30 AM			-11 cr

Figure 2: Credit balance information page

Mobile-Friendly Views (Converting Top Navigation Bar into a Hamburger Menu)

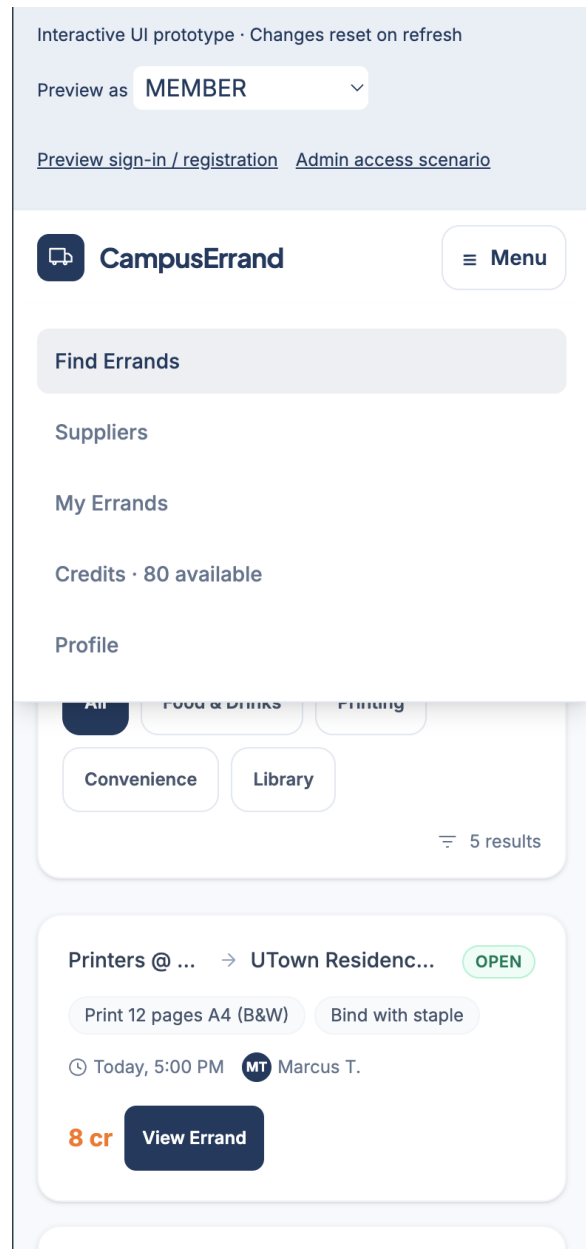


Figure 1: Hamburger menu for page navigation

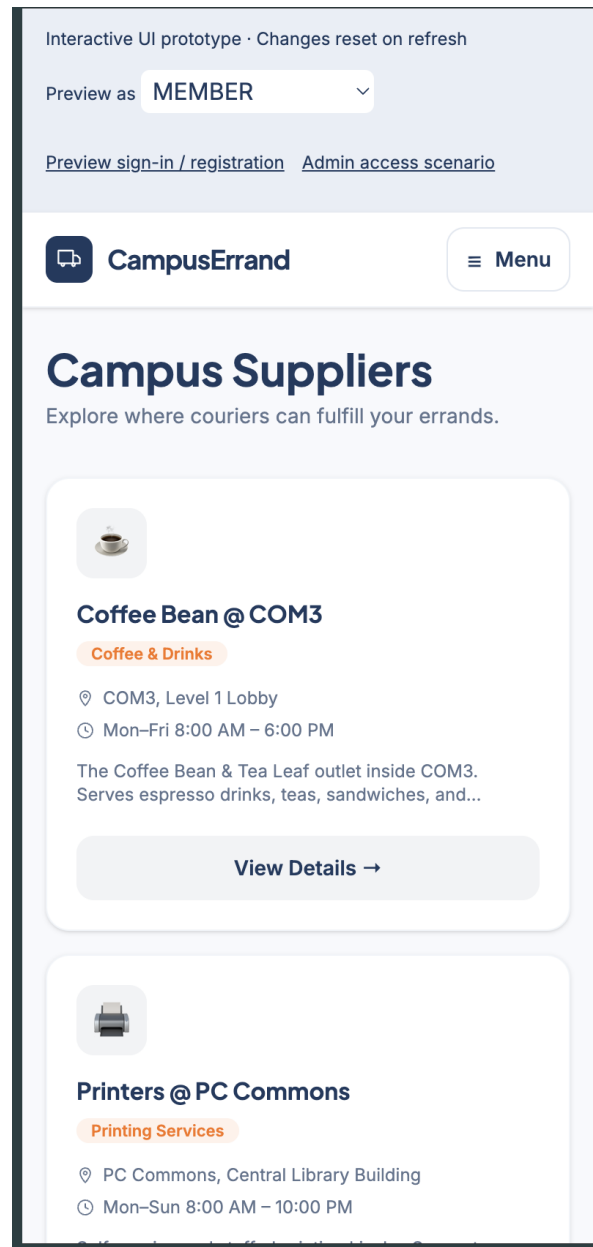


Figure 2: Single-column stacked layout

4. Project Plan

Excel Sheet here (To be screenshotted and placed here)

5. AI Declaration Usage

Tool	Prompt	Response	Purpose / Scope
GPT-5.6-sol-medium	Based of our product requirements, I want to generate a use Figma Make to generate a UI Mockup to envision what the end product could look like, and will thus need help with generating a prompt to be passed into Figma Make. It should follow a clean style that follows desktop first principles be marketplace-themed. It should feature a nav bar at the top with the relevant main buttons leading to different pages such as "Find Errands", "Credits", "Suppliers". For page such as find errands, it should feature a list of mock errands that displays relevant information such as errand status, errand description, errand destination. The other pages should follow similar format based on tis structure.	Paste this into Figma Make: Create a polished clickable web app mockup for a university campus errand-sharing platform called CampusErrand. Students can both request errands and act as couriers using the same account. Style: clean modern university marketplace, desktop-first 1440px, responsive, white/light-grey background, navy primary colour, orange accents, rounded cards, subtle shadows, clear typography, Lucide-style icons. Keep it realistic and implementation-friendly. Navigation: Logo Find Errands Suppliers My Errands Credits Profile, with current credit balance visible. Create these connected screens: 1. Find Errands / Home Header "Errands around campus". Search/filter by pickup, destination, credits and deadline. Show errand cards with pickup → destination, items, reward, deadline, requester, OPEN badge and "View Errand". Add "+ Request an Errand". 2. Create Errand Fields: supplier/pickup location, items, delivery location, instructions, offered credits, listing deadline, pickup deadline, delivery deadline. Show available balance and reserved credits. CTA: "Post Errand". 3. Errand Details	Generate Prompt for Figma Make to prototype a UI

		Show requester, pickup, items, destination, instructions, reward and deadlines. Main CTA: “Accept Errand — Earn 12 Credits”. Include confirmation modal. 4. My Errands Tabs: Requested by Me Delivering Completed. Use lifecycle badges: OPEN, ACCEPTED, PICKED UP, DELIVERED, CANCELLED, EXPIRED, FAILED, DISPUTED. Show progress tracker: Open → Accepted → Picked Up → Delivered. 5. Suppliers Cards for Coffee Bean @ COM3, Printers @ PC Commons, PGP Foyer. Show category, location, opening hours and description. Supplier detail page with “Request Errand From Here”. 6. Credits Summary cards: Total Balance 120, Reserved 24, Available 96. Add transaction history for earned/spent credits. Credits cannot be purchased or withdrawn. Project-D1-Template.pdfPDF Add realistic empty states, errors, confirmation dialogs and toast notifications. Make all screens part of one clickable prototype.	
Figma Make	Create a polished clickable web app mockup for a university campus errand-sharing platform called CampusErrand. Students can both request errands and act as couriers using the same account. Style: clean modern university marketplace, desktop-first 1440px, responsive, white/light-grey background, navy primary colour, orange accents, rounded cards, subtle shadows, clear typography, Lucide-style icons. Keep it realistic and implementation-friendly.	The CampusErrand prototype is live. Here's what's built: 6 fully connected screens: Find Errands — search bar, category filters (Food & Drinks, Printing, Convenience, Library), live result count, errand cards with pickup → destination, items, reward, deadline, requester avatar, and “View Errand” Create Errand — all fields (supplier autocomplete from real supplier list, items, destination, instructions, credits with balance	Generate UI Mockup based off requirements

	Navigation: Logo Find Errands Suppliers My Errands Credits Profile, with current credit balance visible. Create these connected screens: 1. Find Errands / Home Header “Errands around campus”. Search/filter by pickup, destination, credits and deadline. Show errand cards with pickup → destination, items, reward, deadline, requester, OPEN badge and “View Errand”. Add “+ Request an Errand”. 2. Create Errand Fields: supplier/pickup location, items, delivery location, instructions, offered credits, listing deadline, pickup deadline, delivery deadline. Show available balance and reserved credits. CTA: “Post Errand”. 3. Errand Details Show requester, pickup, items, destination, instructions, reward and deadlines. Main CTA: “Accept Errand — Earn 12 Credits”. Include confirmation modal. 4. My Errands Tabs: Requested by Me Delivering Completed. Use lifecycle badges: OPEN, ACCEPTED, PICKED UP, DELIVERED, CANCELLED, EXPIRED, FAILED, DISPUTED. Show progress tracker: Open → Accepted → Picked Up → Delivered. 5. Suppliers Cards for Coffee Bean @ COM3, Printers @ PC Commons, PGP Foyer. Show category, location, opening hours and description. Supplier detail page with “Request Errand From Here”. 6. Credits Summary cards: Total Balance 120, Reserved 24, Available 96. Add transaction history for earned/spent credits. Credits cannot be purchased or withdrawn. Project-D1-Template.pdfPDF Add realistic empty states, errors, confirmation dialogs and toast notifications. Make all screens part of one clickable prototype.	validation, 3 deadline pickers), with a live credit reservation warning and "Post Errand" CTA Errand Details — full route visualisation with dashed connector, items, instructions in an amber callout, requester info, 3 deadline tiles, and "Accept Errand — Earn X Credits" CTA My Errands — 3 tabs (Requested / Delivering / Completed) with lifecycle badges and a step-by-step progress tracker (Open → Accepted → Picked Up → Delivered), plus cancel and mark-delivered actions Suppliers — 4 supplier cards with icon, category, location, hours; supplier detail page with popular items list and "Request Errand From Here" (pre-fills the Create form) Credits — 3 summary cards (Total 120 / Reserved 24 / Available 96), earned/spent stats, platform notice (no purchase/withdrawal), transaction history with filter tabs Interactive elements: accept confirmation modal, cancel errand modal, toast notifications (success/error/info), empty states with contextual CTAs, search + category filtering, and cross-screen navigation with back links.	
Codex-gpt-5.6-sol-medium	We have updated the errand states to “OPEN”, “ACCEPTED”, “PICKEDUP”,	> Partially—the mockup shows the main member journey, but several	Updated states and state transition for order service, so

	"DELIVERED", "COMPLETD", and FR for the order service, is our UI mockup still in sync, if not modify the states to reflect these changes.	requirements are missing, and the delivery lifecycle > differs from your specification. I checked the UI source; this is a coverage review, not a browser test. No files > were changed. > The biggest issue is the delivery and completion distinction: > - Your requirements say: OPEN → ACCEPTED → PICKED_UP → DELIVERED → COMPLETED. > - The mockup's progress tracker ends at DELIVERED, and the Completed tab contains errands labelled DELIVERED. > - Clicking Mark Delivered says "Delivery confirmed! 13 credits earned." Your requirements allow payment only after > requester confirmation or an administrator ruling. > - There are no Mark Picked Up, Withdraw, Confirm Receipt, Reject Delivery, or Resolve Dispute actions. > - Requesters can cancel only OPEN errands; your requirements also allow cancellation while ACCEPTED. > - Another user's errand can show Accept Errand even when it is already picked up or delivered, because the button > checks the requester but not the errand state. > - These are visible mismatches in App.tsx, even for a mockup. > - A few credit labels also need adjusting: > - "Total Balance: All-time accumulated credits" should describe the current balance, including reserved credits. > - "Reserved: Held for open errand requests" is too narrow; credits remain held through later stages and disputes. > - "Credits can only be earned by completing deliveries" should acknowledge the initial 100-credit registration > grant.	we wanted to make sure that the UI mockup reflected that
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